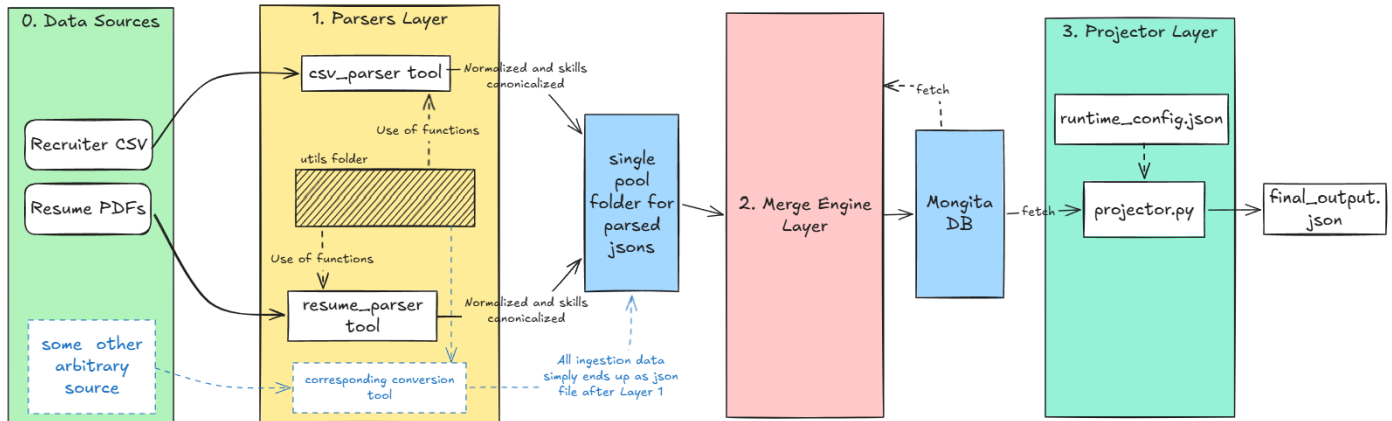


DESIGN DOCUMENT

➤ Pipeline: [Excalidraw URL](#)



➤ Canonical output schemas: Majorly matches the base given in PS. Just added projects field.

➤ Merge/Conflict-resolution policy:

- Skills Canonicalization (In Layer 1): Uses predefined 'groupings' and 'reverse_mapping' for fast canonical keyword finding. Unrecognized skills are isolated to an exception file, allowing an offline, LLM-assisted script to manually update the canonical mappings as new skills emerge. To ensure determinism, the existing skills also have to be re-canonicalized according to new groupings.
- Entity Resolution (Match Keys): Deduplicates profiles based on intersecting either emails or phones. This is robust. Because, it is logically implemented as a Disjoint Set Union (DSU) graph, where nodes represent normalized JSON profiles and edges denote shared contact identifiers.
- Merge Policy & Confidence Scoring: Matched candidate records are resolved field-by-field (e.g., strict union for arrays, single-winner selection for strings). The overall_confidence is calculated dynamically during the merge process by applying predefined weights to each individual field's confidence score.

➤ Runtime Config & Extensibilit:

- Projection & Validation: A decoupled projector applies dynamic JSON rules (filtering, renaming, on_missing logic) and strictly validates the output (via Pydantic)
- NoSQL Extensibility: Staging profiles in Mongita allows developers to bypass basic configs and extract highly custom data shapes using native MongoDB queries.
- LLM Integration: Standardizing on MongoDB syntax enables LLMs to instantly translate natural language requests into complex, increasing breadth of custom configs.

➤ Few Edge cases:

- I assume that all emails are from non-recycling providers (Google, Microsoft, unlike Yahoo).
- I acknowledge that phone numbers are recycled. But, due to time constraints I ignore that.
- Experience date format in source files. Date format assumptions. These are rare formats, but handled using assumptions.
 - XX-XX-XXXX assumed to be DD-MM-YYYY.
 - XXXX-XX-XX assumed to be YYYY-MM-DD [ISO 8601]
 - XX-XXXX-XX assumed to be DD-YYYY-MM